

Thevenin's Theorem

Video 7

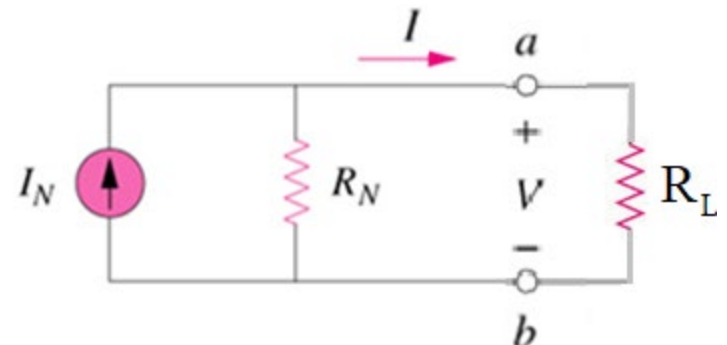
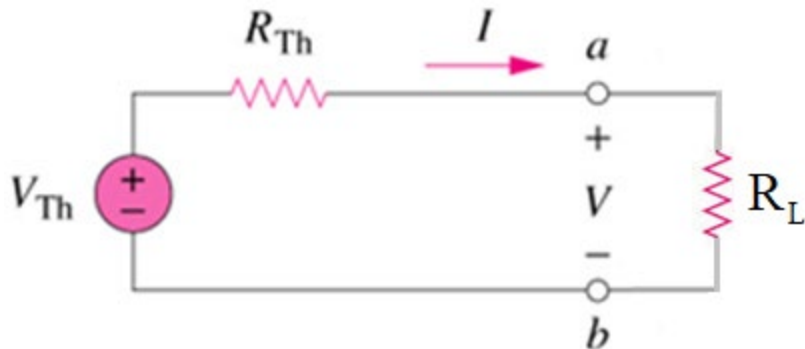
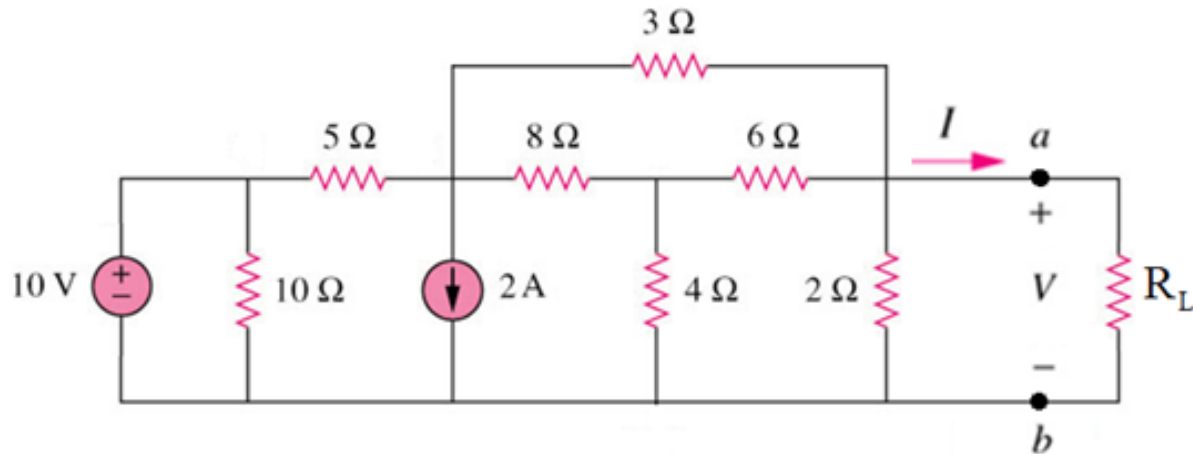
Electrical Science

Dr. Mohamed Bingabr

University of Central Oklahoma

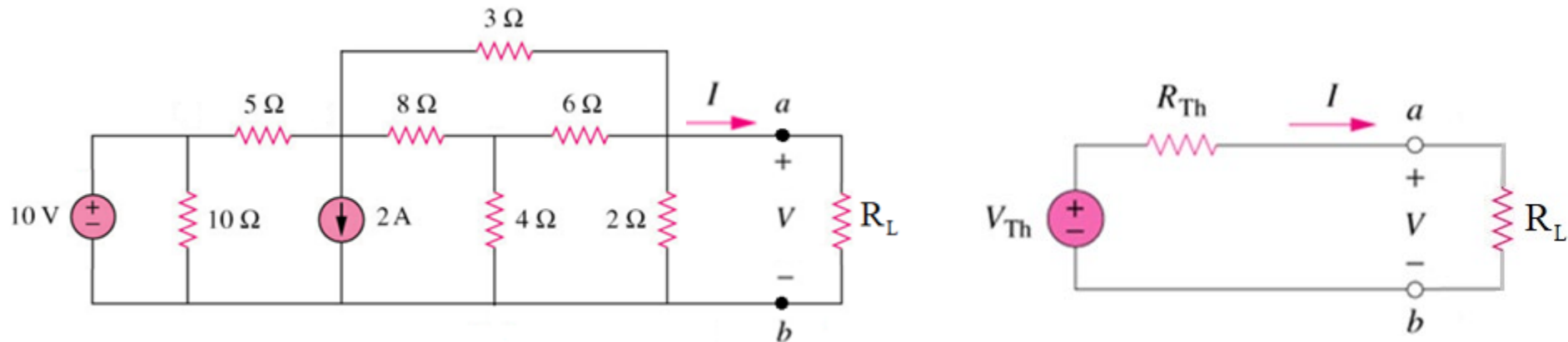
Thevenin's Theorem

- Thevenin's Theorem
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- Maximum Power Transfer



Thevenin's Theorem

It states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source V_{TH} in series with a resistor R_{TH} ,

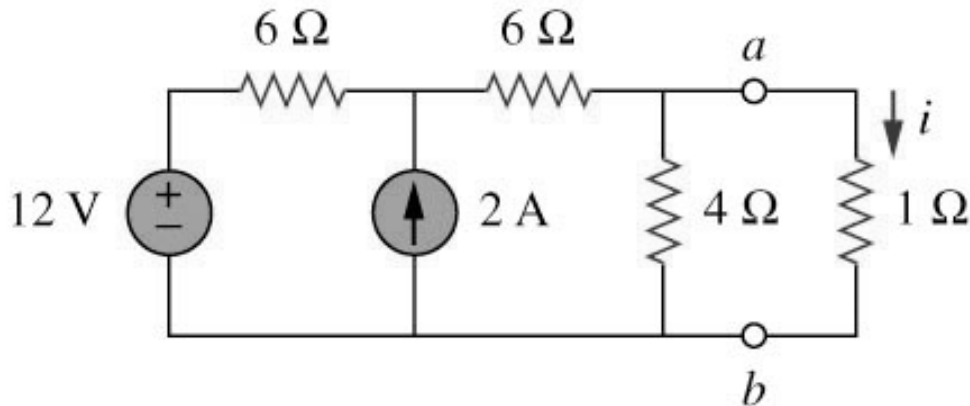


- V_{TH} is the open-circuit voltage at the terminals.
- R_{TH} is the input or equivalent resistance at the terminals when the independent sources are turned off.
- Independent sources are turned off by replacing a Voltage source by a short circuit and current source by open circuit.

Thevenin's Theorem

Example 1

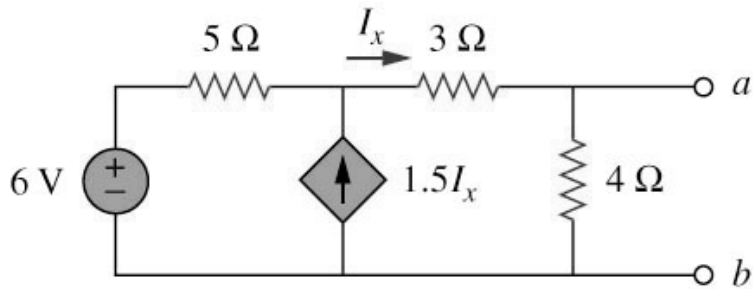
Using Thevenin's theorem, find the equivalent circuit to the left of the terminals in the circuit shown below. Hence find i .



Thevenin's Theorem with dependent source

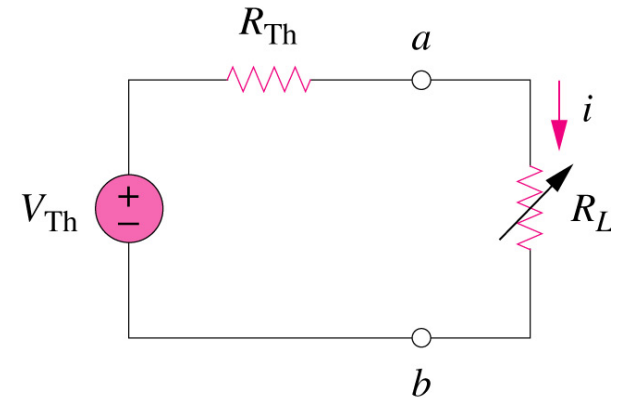
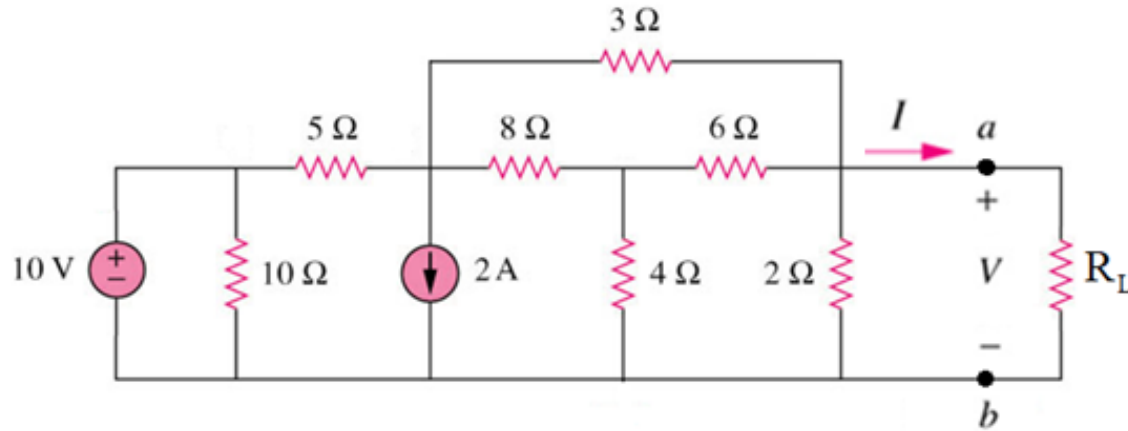
Example 2

Find the Thevenin equivalent circuit of the circuit shown below to the left of the terminals.



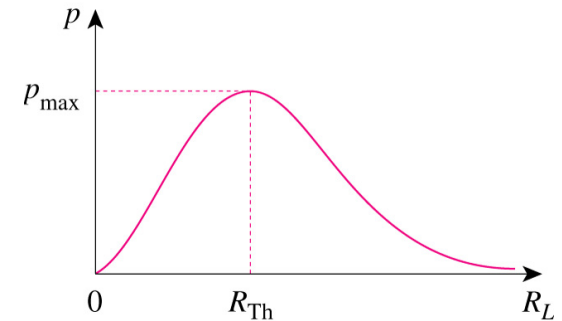
Maximum Power Transfer

What should the value of R_L be to receive maximum power?



If the entire circuit is replaced by its Thevenin equivalent except for the load, the power delivered to the load is:

$$P = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L$$



For maximum power dissipated in R_L , for a given R_{TH} , and V_{TH} ,

$$R_L = R_{TH} \Rightarrow P_{\max} = \frac{V_{Th}^2}{4R_L}$$

Maximum Power Transfer

Example 3

Determine the value of R_L that will draw the maximum power from the rest of the circuit shown below. Calculate the maximum power.

